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A Highly Backdrivable Robotic Arm using Low Friction and High Accuracy Cycloidal Geared Motors: ALFHA

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***Abstract*— This article proposes a robotic arm using low friction and high accuracy geared motors. Reducers installed in conventional industrial robots are designed to have a low backlash of approximately 3 arcmin (0.05 degree) or less to suppress lost motion. However, the joints of the robot are stiff, and thus it is difficult to backdrive by a small external force. This problem limits applications of existing industrial robots, particularly for tasks that require mechanical contact between the robot and the external environment. Although a low friction geared motor constitutes a possible solution, backlash leads to an increase in lost motion. Therefore, this study presents a combination of electric motor and cycloidal drive with a 20:1 reduction ratio, 1 arcmin of backlash, and low backdrive torque (low friction) of 1 Nm. Experiments verify position and orientation repeatability, as well as force accuracy, estimated from the motor-current based joint torque.**

I. INTRODUCTION

Many conventional industrial robots are comprised of electric motors and strain wave gearing (known as harmonic drive [1]), cycloidal drive [2], and planocentric reducers [3]. The torque can be increased due to a high reduction ratio in one stage. These reducers are designed to have a backlash of approximately 3 arcmin (0.05 degree) or less to suppress the lost motion of the robot, which is miniscule. A combination of electric motors and high reduction ratio gears is a pragmatic solution.

However, the joints of the robot are stiff due to an increase in static friction, making it challenging to move by a small external force. Furthermore, to guarantee the position accuracy of the end-effector by controlling the motors in a joint angle coordinate system, the rigidity of the entire structure, such as links, should be augmented. As a result, the robot becomes increasingly rigid against external force.

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If the robot allows active mechanical contact with the environment, the following advantages are obtained:

1) Work with geometric constraint conditions, such as peg-in-hall, can be easily realized.
2) It is possible to prevent damage to the robot and the environment when there is an error in the geometric information of the environment.
3) High-precision work can be achieved by using mechanical constraints. For example, a straight line can be drawn using a ruler, just as humans do.
4) It is possible to realize collaborative work corresponding to the force given by humans and other machines. This is especially useful when multiple robots perform collaborative work involving contact with each other.
5) The characteristics of the object and the environment can be recognized by contact, for example, by applying force, structural coordination, viscoelastic property recognition, etc.

II. RELATED WORKS

Robots with high reduction ratio gears, such as strain wave gearing, have small backlash but inherently stiff joints. Therefore, force/torque sensor feedback control is employed to make them more flexible [4]–[6]. In this method, flexibility can be artificially adjusted as needed, which is well known as impedance control. Indeed, many startup and traditional robot companies have already sold robots based on this method, as it makes effective use of the advantages of current industrial robots [7]–[11]. However, certain limitations exist because the force/torque sensor is expensive and easily damaged, and there is a response delay in the sensor feedback.

On the other hand, the problem of stiff joints in high reduction ratio geared motors is solved by installing elastic materials in series. The flexibility is determined by the properties of the elastic material, and thus it is not variable stiffness. The structure is both simple and unbreakable, however, and there is no response delay of sensor feedback. This solution is the Series Elastic Actuator (SEA) [12], and certain startup companies have adopted it [13]–[18]. SEA tends to induce vibration of the arm end when the robot repeats moving and stopping at high speed. Therefore, the stiffness of the recent most SEA is designed to be high.

If friction, which constitutes an uncertain physical phenomenon, can be eliminated, the motion of the robot can be specified by electromagnetic and mechanical models. To remove the influence of friction in the drive system, a direct drive (DD) system was developed [19], [20]. Robot

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arms using only DD motors have also been researched [21]. DD offers major advantages, such as low friction and easy modeling of robot dynamics. However, to generate sufficient torque, it leads to an increase in size and weight of each joint, and the torque ripple cannot be reduced by using a reduction gear. Some companies still market DD motors [1], [22]–[24].

III. Low Frictional Geared Electric Motor

In order to solve the problems of high weight and torque ripple of DD motors, the combination of low friction gears and electric motors has been investigated. Empirically, it has been demonstrated that gears with a reduction ratio of 10 or less are backdrivable. However, even for high reduction ratios of 100 or more, if backlash is allowed by devising a gear shape or increasing the distance between the shafts, friction in the sliding parts is reduced, and it remains backdrivable. Therefore, the reduction ratio alone does not directly affect backdrivability.

For instance, a compound planetary gear reducer, with a high reduction ratio of approximately 96 and high backdrivability by carefully considering the internal structure of the reducer, was developed [25]. In addition, tiny servo units, called Dynamixel, are designed to backdrive even at reduction ratios of approximately 200:1 to 400:1 [26].

On the other hand, research and development of robots that allow mechanical contact seems to be more active in the field of walking robots. A combination of DD motor and low reduction gear, named the "proprioceptive actuator", which can mitigate the impact applied to the leg of a quadruped walking robot, has been proposed [27]. This actuator is a flat, low-friction geared motor that incorporates a planetary gear reducer with a reduction ratio of 6 to 1 inside of an outer rotor motor with relatively high torque. It can be assumed that this idea is also applied to SPOT developed by the company Boston Dynamics [28].

This method has also been used in robotic arms. For example, BLUE, a 7-DOF twin-arm robot made from a 3D printer, was developed [29]. To eliminate backlash, a single-stage timing belt transmission mechanism with a reduction ratio of approximately 7 was used, which not only ensures flexibility and positioning accuracy when the fingers come into contact with the object, but also eliminates the lack of torque caused by the redundant drive. In addition, the company INNFOS has announced 6-DOF robot arms composed of motor units equipped with a two-stage planetary gear reducer with a reduction ratio of 36 [30]. They gave a talk in the industrial forum of the IROS 2019 in Macau, China.

In recent years, a trend has existed to refer to geared motors with a small reduction ratio and easy backdrive as quasi-DD motors or low reduction ratio geared motors. It is true that the smaller is the reduction ratio, the smaller is the effect of the motor's inertia and viscosity from the output shaft side during backdrive. However, as mentioned previously, there are cases in which backdrive is possible even with a reduction ratio of several hundred. The frictional force of the sliding parts is essentially more dominant than the reduction ratio. Therefore, we use the term "low frictional geared electric motor" in this paper.

Table I presents a comparison of existing methods for mechanical contact of robots. Indeed, low frictional geared electric motors tend to be backdrivable and can be designed in a simple manner. However, previous works with a high backdrivable system [25], [26] sacrificed positioning accuracy (joint backlash is 15 arcmin or more). In addition, quantitative evaluation of backdrivable torque and positioning accuracy has not yet been discussed. Even though some newly-emerging companies claim that the positioning accuracy of their products is ± 0.1 mm, no evidence is shown. Therefore, in this paper, a new robotic arm with high backdrivability and a low backlash geared system based on a cycloidal drive is presented. Positioning repeatability and accuracy of force and moment applied to the arm end are assessed by experiments. Force and moment are calculated by joint torque obtained by built-in current sensors.

IV. Mechanical Design and Control System

Some existing hobby-use or research-use robotic arms are composed of uniformed motor units. However, the required joint torque gradually decreases as it approaches the arm end. Installing a heavy and large motor unit that generates more torque and speed than is necessary at the arm end is unreasonable. Consequently, different structures are designed for each joint. Figure 1 shows the cross-sectional views of three developed motor units. To drive the joint, BLDC (Burushless Direct Current) motors of TQ-RoboDrive (TQ-Systems GmbH, Gut Delling, Mühlstraße 2 82229 Seefeld Germany) are installed in each joint.

For joint 0 and 1 (motor unit A), TQ-RoboDrive ILM70x10 is used. The rotor angle and the output angle are measured by Renishaw MB029-F and MB049-D (both manufactured by Renishaw plc, New Mills, Wotton-under-Edge, Gloucestershire, GL12 8JR, UK), respectively. For joint 2 (motor unit B), TQ-RoboDrive ILM50x08 is selected, and both rotor and output angles are measured by the same Renishaw encoder as that used in joint 0 and 1 (MB029-F). The above three motor units have the same cycloidal drive with a 20:1 of reduction ratio. For joint 3 (motor unit C), TQ-RoboDrive ILM25x08 is adopted to reduce the size and weight. Renishaw RMB20SC and AS5048A (ams AG, Tobelbader Strasse 30, 8141 Premstaetten, Austria) are installed for rotor angle measurement and output angle measurement, respectively. Unlike joint 0, 1, and 2, a different cycloidal drive is mounted, although the reduction ratio is same as that of the other joints.

In our study, to achieve high backdrivability, a low friction reducer based on a cycloidal drive was developed. To suppress the rotor inertia and viscosity, 20:1 of the reduction ratio and inner rotor type motor were selected, even though the general backdrivable actuator is the outer rotor type [27].

Although detailed information about this gear cannot be currently published for reasons of confidentiality, evaluation as a torque sensor-less robotic arm can be performed.

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TABLE I

COMPARISON AMONG EXISTING METHODS FOR MECHANICAL CONTACT OF ROBOTS

Performance	Positioning repeatability	Impact resistance	Size and weight	Additional sensor
Force/torque sensor feedback control	High	Weak	Large and heavy	Force/torque sensor
SEA (Series Elastic Actuator)	Low	Tough	Large and moderate	Two position sensors
Low friction geared motor	Depends on backlash	Tough	Small and light	None

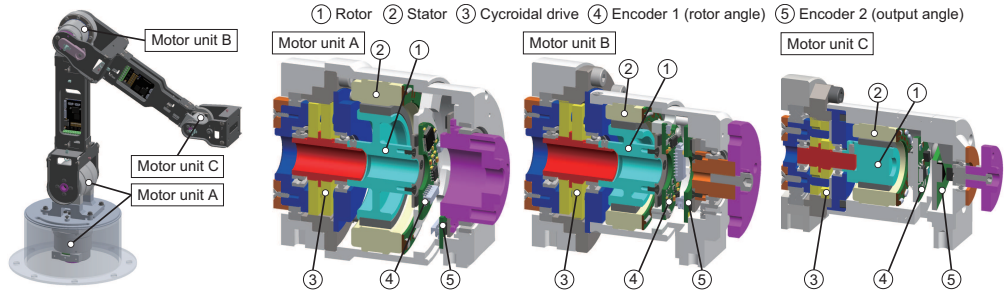


Fig. 1. Motor units installed in each joint: (a) inner skeleton overview of the 4-DoF robotic arm, motor units of (b) joints 0 and 1, (c) joint 2, and (d) joint 3 (arm end).

TABLE II

SPECIFICATION OF THE ROBOTIC ARM (ALFHA)

Robot weight [kg]	5.5
Total length [m]	0.869
Max. end point speed [m/s]	2.1
Weight capacity [kg]	2.5

TABLE III

SPECIFICATIONS OF EACH MOTOR UNIT

Joint number (motor unit)	1 & 2 (A)	3 (B)	4 (C)
Reduction ratio	20:1		
Rated torque [Nm]	13.2	5.96	1.26
Peak torque [Nm]	42.6	19.2	4.08
Approx. backlash [arcmin]	1	1	3
Approx. backdrive torque [Nm]	1	1	1
Approx. weight [kg]	1.0	0.77	0.28
Maximum axial length [m]	0.09	0.09	0.06
Maximum diameter [m]	0.08	0.08	0.04

Tables II and III list the specifications of the developed robotic arm and the three developed motor units, respectively. The rated and peak torque are both theoretical values. Considering efficiency of the reducer, the true value might be smaller than the given value (generally 70 % of the theoretical value).

Figure 2 illustrates the control system of our proposed robotic arm. For the motor control, SOMANET Node 400 (Synapticon GmbH, Daimlerstraße 26 71101 Schünaich Germany) is mounted on the robot. The simple PID position control law, originally embedded in SOMANET, was adopted. EtherCAT (Ethernet for Control Automation Technology) was used to send a command and to receive a signal from the robot. For high speed communication, PCI express card

TABLE IV

FEEDBACK GAINS IN SOFT AND RIGID MODES

Soft mode				
Joint number	Joint 0	Joint 1	Joint 2	Joint 3
Proportional gain [Nm/deg]	0.44	0.58	0.29	0.18
Integral gain [Nm/degs]	5.83	5.83	2.91	0.91
Derivative gain [Nms/deg]	0.175	0.117	0.058	0.055
Rigid mode				
Joint number	Joint 0	Joint 1	Joint 2	Joint 3
Proportional gain [Nm/deg]	8.74	14.56	14.56	1.82
Integral gain [Nm/degs]	29.13	43.69	43.69	6.37
Derivative gain [Nms/deg]	0.029	0.058	0.058	0.005

DPE-3011 (DENSAN Co. Ltd., 1-25-16 Kamitakaido, Suginami, Tokyo, Japan) was installed. Windows 10 and Visual Studio 2019 were used for IDE (Integrated Development Environment).

Our developed robotic arm can achieve variable stiffness by tuning the feedback gain of the controller. Tables IV lists the gain values in soft mode and rigid mode, respectively. Theoretically, feedback gains can be reduced further and set to zero. The robotic arm, however, cannot support its own weight with low feedback gains. On the other hand, the gains cannot be increased infinitely. This is because the joint torque against the angular displacement reaches its limit. To increase the stiffness, a higher power motor needs to be selected, but this will lead to increase in weight and size. The feedback gains given in the rigid mode constitute the limitations of our developed robot.

V. EXPERIMENTS

A. Trajectory planning

To make the end of the robotic arm smoothly trace the target trajectory, the target position is given by the cubic

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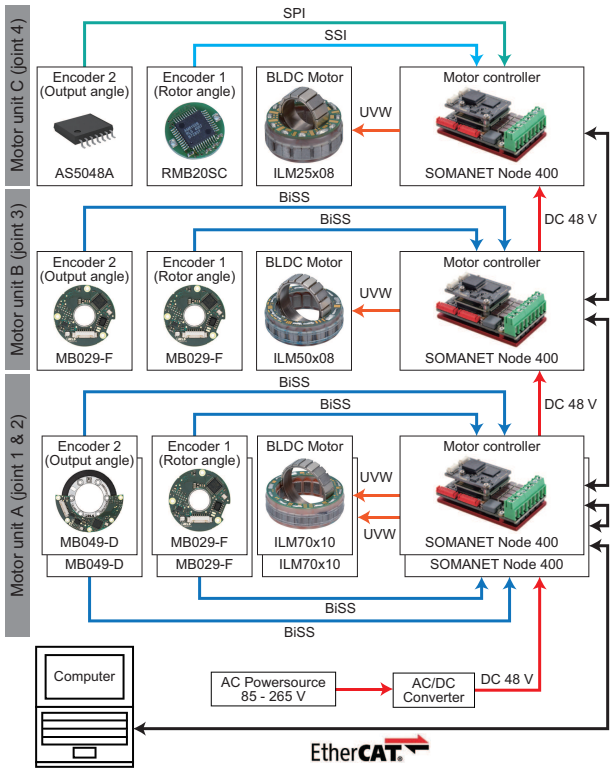


Fig. 2. Control system of the robotic arm.

polynomial function with respect to time (t) as follows:

$$\boldsymbol{r_d} = \left(-\frac{2t^3}{T^3} + \frac{3t^2}{T^2}\right)(\boldsymbol{r_T} - \boldsymbol{r_{ini}}) + \boldsymbol{r_{ini}} \tag{1}$$

$\boldsymbol{r_d}$, $\boldsymbol{r_T}$, and $\boldsymbol{r_{ini}}$ denote the target positions and orientation of the arm end at a time point t, the positions and orientation of the arm end at $t = T$ (entire control time), the initial positions and orientation of the arm end, respectively, which are defined by $\boldsymbol{r_d} = \begin{bmatrix} x_\mathrm{d} & y_\mathrm{d} & z_\mathrm{d} & \phi_\mathrm{d} \end{bmatrix}^\mathrm{T}$, $\boldsymbol{r_T} = \begin{bmatrix} x_T & y_T & z_T & \phi_T \end{bmatrix}^\mathrm{T}$, and $\boldsymbol{r_{ini}} = \begin{bmatrix} x_\mathrm{ini} & y_\mathrm{ini} & z_\mathrm{ini} & \phi_\mathrm{ini} \end{bmatrix}^\mathrm{T}$.

The calculated target positions and orientations are converted to target joint angles via inverse kinematics. Each link length is designed to L_0=0.1738 m, L_1=0.3 m, L_2=0.3 m, and L_3=0.0952 m. The initial positions and orientations are obtained from the encoder values and forward kinematics. The feedback gains were selected from the combination of the rigid mode listed in Table IV.

B. Position/posture accuracy measurement

First, the position and posture of the developed robotic arm are evaluated. Two markers were attached to the arm end to measure positioning repeatability, as depicted in Fig. 3. Center position and orientation of the marker were captured by two area scan cameras acA2500-14um (Basler AG, An der Strusbek 60-62, 22926 Ahrensburg, Germany) and image processing. Resolution of the camera is x_c: 0.05 mm/pixel; y_c: 0.05 mm/pixel; and ψ (orientation): 0.057 deg. Note that

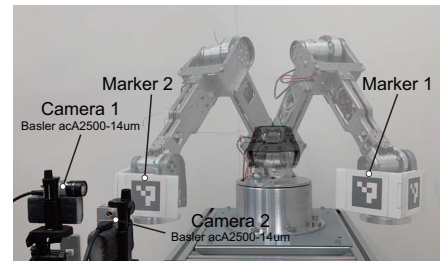


Fig. 3. Experimental setup for positioning repeatability measurement.

TABLE V
RESULTING POSITIONING REPEATABILITY OF THE ARM END

Camera 1 resolution: x_{c1}=0.05 mm/pixel, y_{c1}=0.05 mm/pixel, ϕ_1=0.057 deg			
	x_{c1} [pixel]	y_{c1} [pixel]	ψ_1 [deg]
Max.	1316.5	959	89.503
Min.	1312.5	954	89.189
Error	$\pm$2 ($\pm$0.1 mm)	$\pm$2.5 ($\pm$0.125 mm)	$\pm$0.1572
Camera 2 resolution: x_{c2}=0.05 mm/pixel, y_{c2}=0.05 mm/pixel, ϕ_2=0.057 deg			
	x_{c2} [pixel]	y_{c2} [pixel]	ψ_2 [deg]
Max.	1361	952.5	91.404
Min.	1359	946.5	91.294
Error	$\pm$1 ($\pm$0.05 mm)	$\pm$3 ($\pm$0.15mm)	$\pm$0.0549

the camera coordinate system is totally different from the robotic arm coordinate system.

In this experiment, a pick-and-place movement, as a typical movement of industrial robotic arms, was replicated. Based on equation (1), $\boldsymbol{r_T}$ is given below:

$$\boldsymbol{r_T} = \begin{cases} \begin{bmatrix} 0.35 & -0.2 & 0.2 & \pi \end{bmatrix}^\mathrm{T} & (t = 0\ \mathrm{s}) \\ \begin{bmatrix} 0.35 & -0.2 & 0.1 & \pi \end{bmatrix}^\mathrm{T} & (t = 1\ \mathrm{s}) \\ \begin{bmatrix} 0.35 & -0.2 & 0.2 & \pi \end{bmatrix}^\mathrm{T} & (t = 2\ \mathrm{s}) \\ \begin{bmatrix} 0.35 & 0.2 & 0.2 & \pi \end{bmatrix}^\mathrm{T} & (t = 3\ \mathrm{s}) \\ \begin{bmatrix} 0.35 & 0.2 & 0.1 & \pi \end{bmatrix}^\mathrm{T} & (t = 4\ \mathrm{s}) \\ \begin{bmatrix} 0.35 & 0.2 & 0.2 & \pi \end{bmatrix}^\mathrm{T} & (t = 5\ \mathrm{s}) \end{cases} \tag{2}$$

This cycle was repeated 10 times in total.

Table V lists the resulting positioning repeatability. Considering the resolution of the camera, positioning repeatability in x_{c1}-, y_{c1}, x_{c2}- and y_{c2}-coordinate can be recalculated as $\pm$0.1 mm, $\pm$0.125 mm, $\pm$0.05 mm, $\pm$0.15 mm, respectively. Since the reducer's backlash is miniscule (less than 1 arcmin) and the encoder resolution is sufficiently high, the error is essentially expected to be smaller than measured. However, the link structure was made by sheet metal and assembled by screws, and thus these backlashes were assumed to lead to the error.

C. Force/moment accuracy measurement

Second, force and moment of the arm end calculated by the joint torque were verified. Figure 4 shows the exper-

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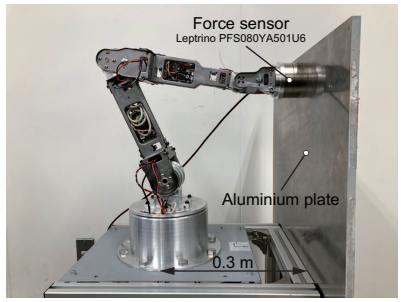


Fig. 4. Experimental setup for force/moment accuracy measurement.

imental setup for the force/moment accuracy measurement. Force sensor Leptrino PFS080YA501U6 (Leptrino Inc., 328-1 Hirahara, Komoro, Nagano, Japan) was attached to the arm end. An aluminum plate was fixed to the base of the robot. The distance between the origin of the global coordinate system and the aluminum plate is 0.3 m.

The trajectory of the arm end $\boldsymbol{r_T}$ is given as follows:

$$\boldsymbol{r_T} = \begin{cases} \begin{bmatrix} 0.25 & 0 & 0.4 & \pi/2 \end{bmatrix}^{\mathrm{T}} & (t = 0\ s) \\ \begin{bmatrix} 0.31 & 0 & 0.4 & \pi/2 \end{bmatrix}^{\mathrm{T}} & (t = 1\ s) \\ \begin{bmatrix} 0.25 & 0 & 0.4 & \pi/2 \end{bmatrix}^{\mathrm{T}} & (t = 2\ s) \\ \begin{bmatrix} 0.31 & 0 & 0.4 & \pi/2 \end{bmatrix}^{\mathrm{T}} & (t = 3\ s) \\ \begin{bmatrix} 0.25 & 0 & 0.4 & \pi/2 \end{bmatrix}^{\mathrm{T}} & (t = 4\ s) \\ \begin{bmatrix} 0.31 & 0 & 0.4 & \pi/2 \end{bmatrix}^{\mathrm{T}} & (t = 5\ s) \end{cases} \tag{3}$$

This cycle was repeated a total of five times. Since the target x-position was set to 0.31 m, the arm end sinks 0.01 m into the aluminum plate. In reality, however, this does not occur because the plate is sufficiently stiff. Instead, joint torques increase, and they depend on the feedback gains. In general, industrial robots with low backdrivability cannot perform the same task using only position control.

Figure 5 plots the resulting target and measured joint angles during the pushing movement. Three waves can be observed when the arm end sinks into the aluminum plate. Target and real angles are not identical because the movement is interrupted by the aluminum plate. Joint 0 to 2 worked well, but joint 3 (arm end) did not follow the target trajectory. This is because the friction of the reducer is much larger than that of the other joints. The joint does not rotate as long as the integral term of the control law (torque) is not accumulated to exceed the static friction. The cycloidal drive used for joint 3 is still under development, and thus it will be refined in the near future.

Figure 6 shows the measured torque of each joint during the pushing movement. At approximately 10 s, the arm end first touched the aluminum plate. At this moment, a spike-like wave was observed in all joints. When the robot does not contact with any external environment, torque was generated to support its own weight. In this experiment, τ_0 should be zero, irrespective of the robot state. However, a slight

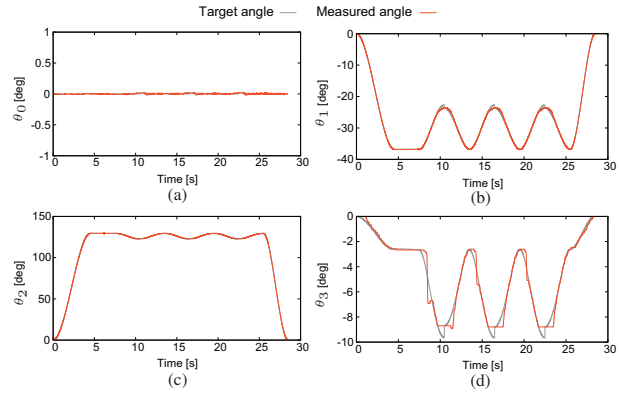


Fig. 5. Target and measured angle of joint 0 (a), joint 1 (b), joint 2 (c), and joint 3 (d) during the pushing movement.

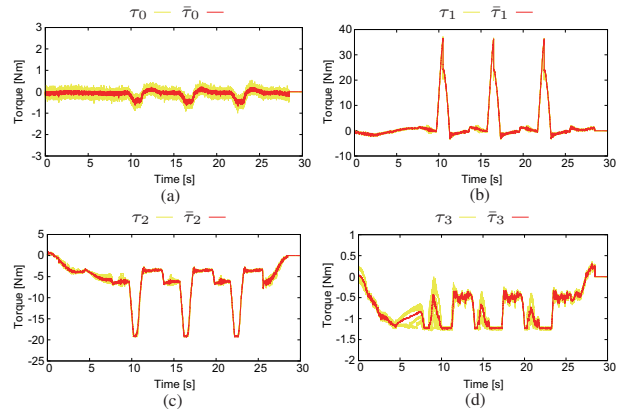


Fig. 6. Measured torque of joint 0 (a), joint 1 (b), joint 2 (c), and joint 3 (d) during the pushing movement.

moment around the z-axis was generated due to the slight orientation error of the arm end. As plotted in Fig. 6 (a), this moment was observed. In the case of the general high friction geared motor, it is balanced and canceled with static friction. It was revealed that 0.1 Nm or less of external moment can be detected with our developed low friction geared motor.

Force and moment applied to the arm end is calculated by:

$$\hat{\boldsymbol{f}} = (\mathbf{J}^{\mathrm{T}})^{-1}\boldsymbol{\tau} \tag{4}$$

where $\hat{\boldsymbol{f}} \in \Re^{4\times1}$ is the vector of external forces and moment; $\mathbf{J} \in \Re^{4\times4}$ is Jacobian matrix; and $\boldsymbol{\tau} \in \Re^{4\times1}$ is the vector of motor torques, which are defined as $\hat{\boldsymbol{f}} = \begin{bmatrix} \hat{f}_x & \hat{f}_y & \hat{f}_z & \hat{m}_y \end{bmatrix}^{\mathrm{T}}$ and $\boldsymbol{\tau} = \begin{bmatrix} \tau_0 & \tau_1 & \tau_2 & \tau_3 \end{bmatrix}^{\mathrm{T}}$ where $\hat{f}_x$, $\hat{f}_y$, $\hat{f}_z$, and $\hat{m}_y$ denote estimated forces and moment calculated from the value shown in Fig. 6 and equation (4); τ_0, τ_1, τ_2, and τ_3 are the torque of each joint.

Figure 7 shows a comparison between the forces measured directly from the force sensor and estimated from the joint torques, where f_x, f_y, f_z, and m_y are forces in x-, y-, and z-directions and moment around the y-axis measured by the force sensor; $\bar{f}_x$, $\bar{f}_y$, $\bar{f}_z$, and $\bar{m}_y$ are the average of f_x, f_y, f_z, and m_y in five trials, respectively; and $\bar{\hat{f}}_x$, $\bar{\hat{f}}_y$, $\bar{\hat{f}}_z$, and $\bar{\hat{m}}_y$ are

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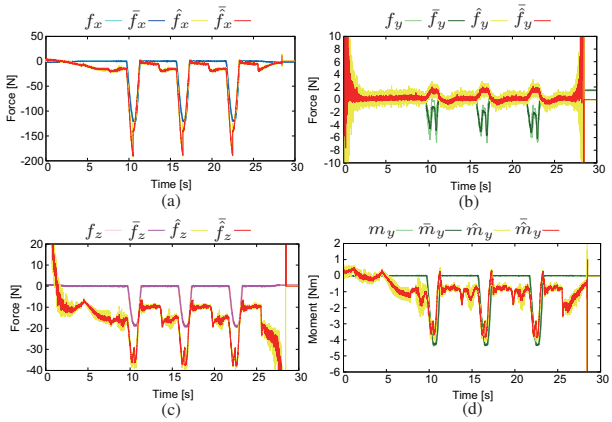


Fig. 7. Measured and estimated end-forces in x-direction (a), y-direction (b), z-direction (c), and moment around y-axis (d) during the pushing movement.

the average of $\hat{f}_x$, $\hat{f}_y$, $\hat{f}_z$, and $\hat{m}_y$ in five trials, respectively.

Since the robotic arm starts movement from a singular configuration, the absolute values of $\hat{f}_y$ and $\hat{f}_z$ were extremely high from 0 to 3 s and from 25 to 28 s, respectively. Ideally, the arm end should slide in only the x-direction because only x_T changed. Therefore, f_y, f_z, and m_y should be zero. However, since the arm end was slightly tilted, f_y, f_z, and m_y were not zero.

From the results, although $\hat{f}_y$ and $\hat{f}_z$ are not identical to f_y and f_z, respectively, the peak value of $\hat{f}_x$ and $\hat{m}_y$ were similar to that of f_x and m_y, respectively. This is because the gear friction of joint 3 is not negligible, and the arm end did not perfectly push the plate in the x-direction. Since the dynamic effect and gravity were not considered, estimated forces and moment were not zero even when the arm end does not touch the plate. Canceling those influences by applying a robot model, it can be eliminated to some extent.

VI. CONCLUSIONS

In the present paper, a robotic arm using low friction and high accuracy geared motors was presented. First, mechanical contact problem that previous robots are currently facing, and typical solutions to this were discussed. Second, the structure and control system of the developed robotic arm were explained. Finally, positioning repeatability and force accuracy estimated from joint torque were examined.

In the future, we will further investigate the force and moment accuracy in which the dynamic effect and gravity are canceled. In addition, the friction of the reducer used for joint 3 remains high, and further refinement is needed. A more detailed position and orientation repeatability test also constitutes a focus of our future work.

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